

Conservation of Energy (Recap)

SS ILP Physics (Intermediate), Week 4 Class 4



Outline

- Conservative Vs Non-conservative Force (Recap)
- Conservation of Mechanical Energy (Recap, with conservative force)
- General Law of Conservation of Energy (Recap, with non-conservative force)

Conservative Vs Non-conservative Force

- A conservative force allows conversion between kinetic and potential energy. Gravitational and the spring force are conservative.
- The work done between two points by any conservative force
 - I. is independent of the path between the two points.
 - II. is zero if the starting and ending points are the same.
 - III. is reversible.
 - IV. can be expressed in terms of a **potential energy function**. Note: only conservative force allows to define potential energy meaningfully due to property (I).
- A force (such as friction, air drag) that is not conservative is called a **non-conservative force**, or a *dissipative force*. Note: not all non-conservative forces are dissipative! e.g magnetic (Lorentz) force.

R.A. Serway and J.W. Jewett, Physics for Scientists and Engineers with Modern Physics. 9E, Cengage Learning, Boston, 2014.

Conservation of Mechanical Energy (Gravitational forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{grav} = -\Delta U_{grav}$$

$$K_2 - K_1 = -(U_{grav,2} - U_{grav,1})$$

$$K_1 + U_{grav,1} = K_2 + U_{grav,2}$$

$$\frac{1}{2}mv_1^2 + mgy_1 = \frac{1}{2}mv_2^2 + mgy_2$$

Total mechanical energy
at position y_1



Total mechanical energy
at position y_2

Conservation of mechanical energy:

When only the gravity does work, the total mechanical energy of the system is conserved:

$$E = K + U_{grav} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Conservation of Mechanical Energy (Spring forces only, Recap)

From work-energy theorem,

$$W_{total} = \Delta K$$

We obtain

$$\Delta K = W_{total} = W_{el} = -\Delta U_{el}$$

$$K_2 - K_1 = -(U_{el,2} - U_{el,1})$$

$$K_1 + U_{el,1} = K_2 + U_{el,2}$$

$$\frac{1}{2}mv_1^2 + \frac{1}{2}kx_1^2 = \frac{1}{2}mv_2^2 + \frac{1}{2}kx_2^2$$

Total mechanical energy
at position x_1



Total mechanical energy
at position x_2

Conservation of mechanical energy:

When only the spring force does work, the total mechanical energy of the system is conserved:

$$E = K + U_{el} = \text{constant}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's
University Physics. 13E, Addison-Wesley, Boston, 2012.

Conservation of Mechanical Energy

- Recall from previous lesson, the total mechanical energy is the summation of kinetic energy and potential energy:

$$E = K + U$$

- In the absence of non-conservative forces (we define this in next section), such as friction, total mechanical energy of an isolated system of objects is constant. It means that kinetic and potential energy simply converts into one another in the system.

$$E_i = E_f \quad \longleftrightarrow \quad K_i + U_i = K_f + U_f$$

- Or, if more than one type of potential energy involved

$$K_i + \sum U_i = K_f + \sum U_f$$

General Law of Conservation of Energy

- Friction and other non-conservative forces remove/add mechanical energy from/to the system. This is the amount by which the mechanical energy of the system changes, denoted by W_{other} . Thus,

$$K_i + U_i + W_{\text{other}} = K_f + U_f \quad \longleftrightarrow \quad E_i + W_{\text{other}} = E_f$$

- Example: In the case of kinetic friction and air resistance, since the force and displacement are always in opposite direction, the work done is always negative, and the final mechanical energy is always less than the initial mechanical energy (where d is distance travelled):

$$K_i + U_i - |\vec{F}_k|d = K_f + U_f \quad \longleftrightarrow \quad E_i - |\vec{F}_k|d = E_f$$

General Law of Conservation of Energy

- Non-conservative force cannot represent by potential energy, but we can describe the effects of these forces in terms of kinds of energy other than kinetic or potential energy.
- Example, when a car with locked brakes skids to a stop, both the tires and the road surface become hotter. The energy associated with this change in the state of the materials is called **Internal Energy** U_{int} .
- Experimentally, we observed that change in internal energy is:

$$\Delta U_{\text{int}} = -W_{\text{other}}$$

H.D. Young and R. A. Freedman, Sears and Zemansky's University Physics. 13E, Addison-Wesley, Boston, 2012.

General Law of Conservation of Energy

- We can write the law of conservation of energy as:

$$K_i + U_i - \Delta U_{\text{int}} = K_f + U_f \quad \longleftrightarrow \quad \Delta E + \Delta U_{\text{int}} = 0$$

Energy can never be created or destroyed. Energy may be transformed from one form to another, but the total energy of an isolated system is always constant!!!

Strategy for applying energy conservation

- Choose initial & final points.
- Select a zero-reference point for each form of P.E., and write the corresponding expression of P.E.
- If the net work done is only due to conservative force(s), apply

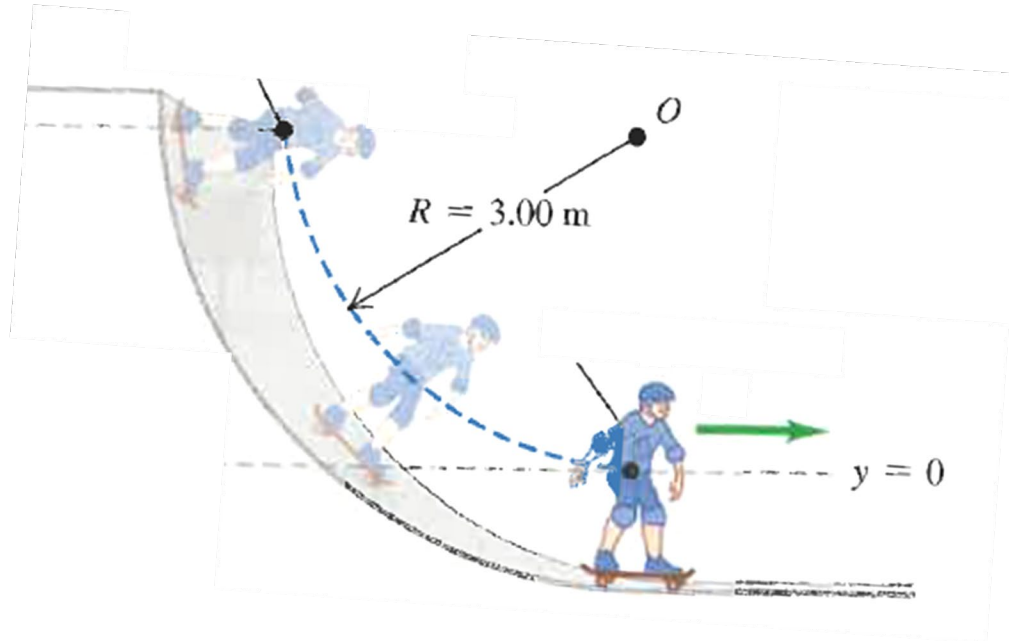
$$\boxed{E_i = E_f} \longleftrightarrow \boxed{K_i + \sum U_i = K_f + \sum U_f}$$

- If the net work done is (partly) due to non-conservative force(s) (like friction, air resistance), apply

$$\boxed{E_i + W_{\text{other}} = E_f \quad \text{or} \quad \Delta E + \Delta U_{\text{int}} = 0}$$

Case Problem 1

Your cousin Throckmorton skateboards from rest down a curved, frictionless ramp. If we treat Throcky and his skateboard as a particle, he moves through a quarter-circle with radius $R=3.00$ m. Throcky and his skateboard have a total mass of 25.0 kg. (a) Find his speed at the bottom of the ramp.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.4, p233

Case Problem 1: Solution

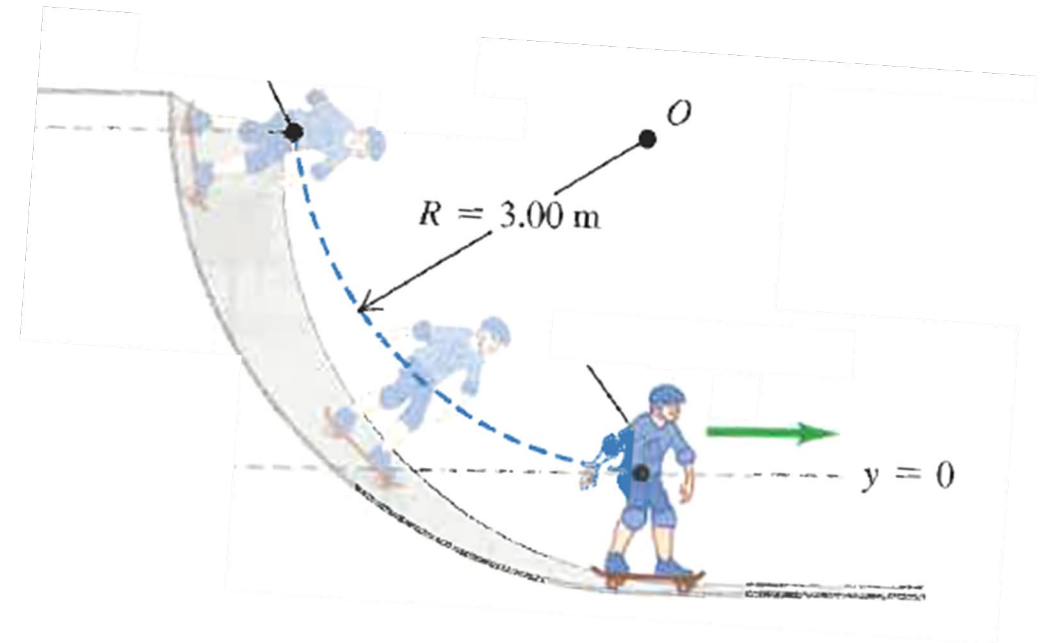
Your cousin Throckmorton skateboards from rest down a curved, frictionless ramp. If we treat Throcky and his skateboard as a particle, he moves through a quarter-circle with radius $R=3.00$ m. Throcky and his skateboard have a total mass of 25.0 kg. Find his speed at the bottom of the ramp. Take $g = 9.8 \text{ m/s}^2$

From conservation of mechanical energy,

$$K_i + U_i = K_f + U_f$$

$$0 + mgR = \frac{1}{2}mv_f^2 + 0$$

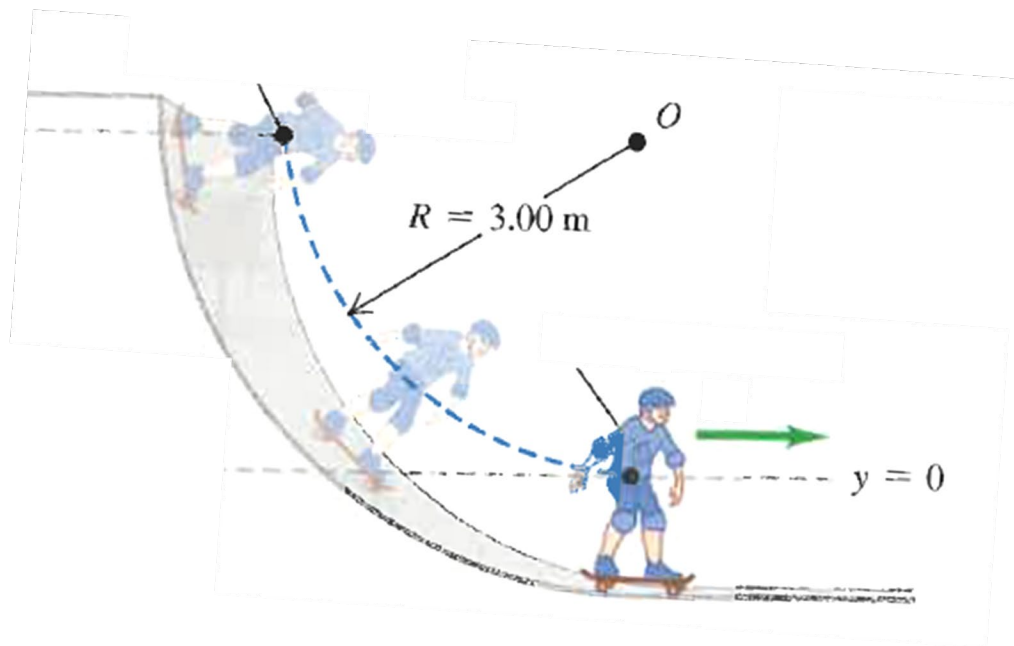
$$v_f = \sqrt{2gR} = 7.67 \text{ m/s}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.4, p233

Case Problem 2

Suppose that the ramp of Case Problem 1 is not frictionless and that Throcky's speed at the bottom is 6.00 m/s. What work was done on him by the friction force?



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.5, p234

Case Problem 2: Solution

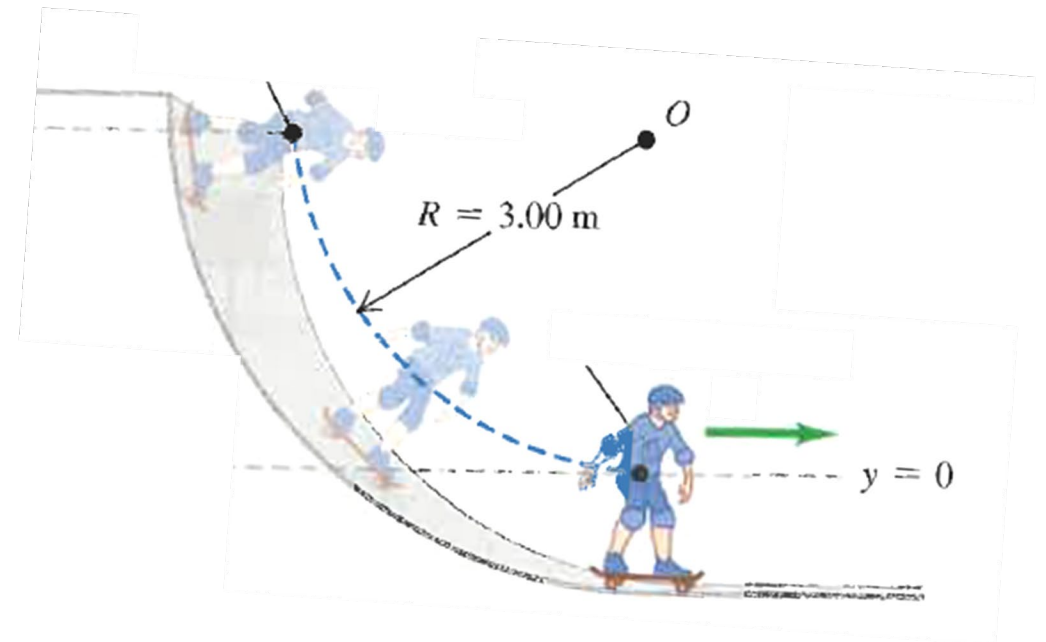
Suppose that the ramp of Case Problem 1 is not frictionless and that Throcky's speed at the bottom is 6.00 m/s. What work was done on him by the friction force?

From conservation of energy,

$$K_i + U_i + W_{other} = K_f + U_f$$

$$0 + mgR + W_f = \frac{1}{2}mv_f^2 + 0$$

$$W_f = \frac{1}{2}mv_f^2 - mgR = -285 \text{ J}$$

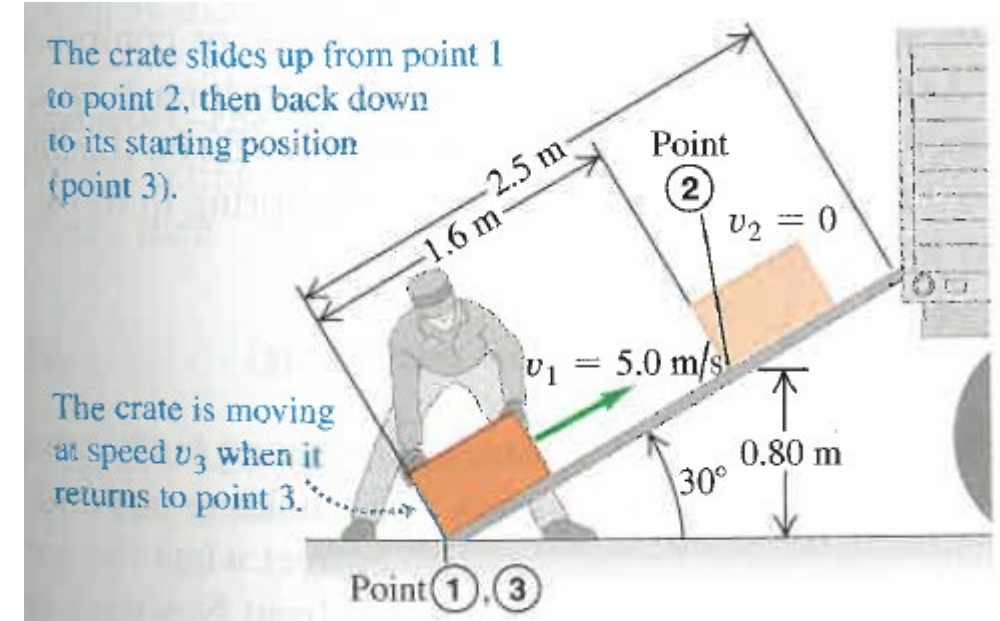


H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.5, p234

Case Problem 3

We want to slide a 12-kg crate up a 2.5-m-long ramp inclined at 30° . A worker, ignoring friction, calculate that he can do this by giving it an initial speed of 5.0 m/s at the bottom and letting it go. But friction is not negligible; the crate slides only 1.6 m up the ramp, stops, and slides down. Take $g = 9.8 \text{ m/s}^2$

- (a) Find the magnitude of the friction force acting on the crate, assuming that it is constant.
- (b) How fast is the crate moving when it reaches the bottom of the ramp.



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.6, p235

Case Problem 3 : Solution

- a) Find the magnitude of the friction force acting on the crate, assuming that it is constant.

From conservation of energy,

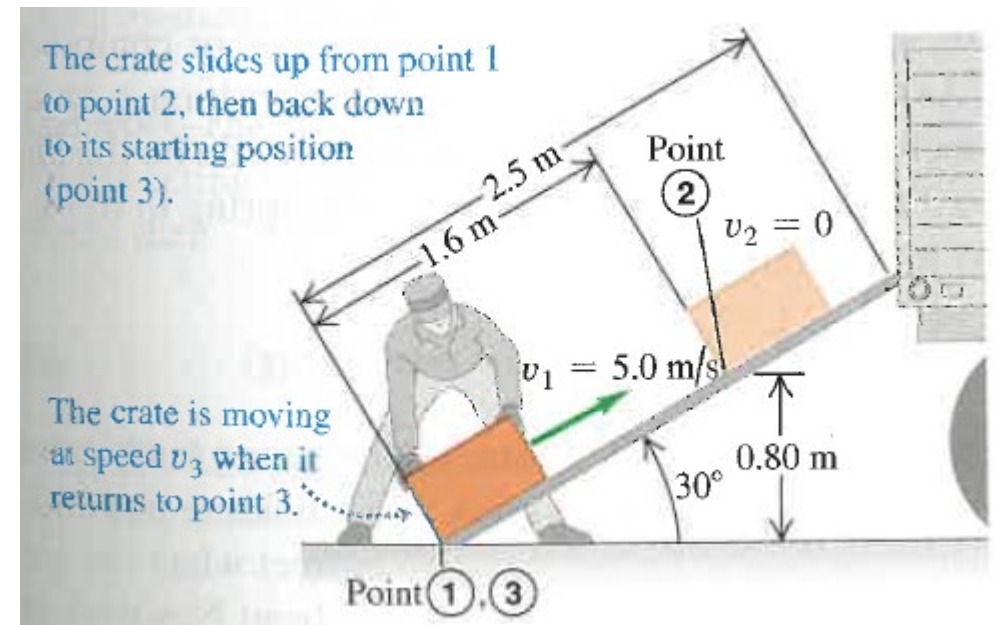
$$K_i + U_i + W_{other} = K_f + U_f$$

$$\frac{1}{2}mv_0^2 + 0 + W_f = 0 + mgh$$

$$W_f = mgh - \frac{1}{2}mv_0^2 = -56 \text{ J}$$

$$W_f = -fs = -56 \text{ J}$$

$$f = 35 \text{ N}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.6, p235

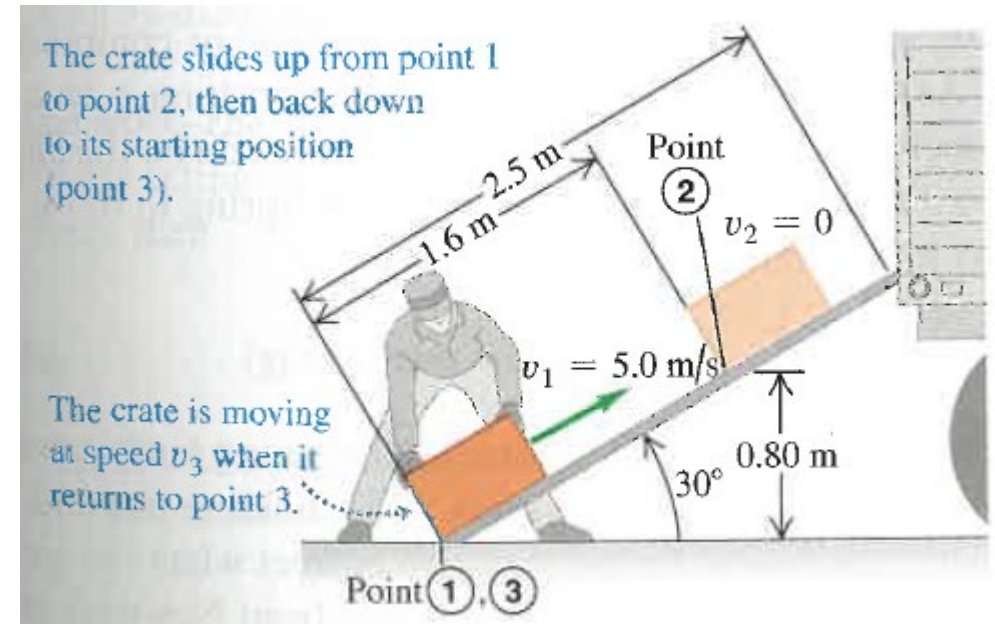
Case Problem 3: Solution

b) How fast is the crate moving when it reaches the bottom of the ramp.

From conservation of energy,

$$K_i + U_i + W_{other} = K_f + U_f$$
$$\frac{1}{2}mv_0^2 + 0 + W_f = \frac{1}{2}mv_f^2 + 0$$
$$\frac{1}{2}mv_f^2 = W_f + \frac{1}{2}mv_0^2 = 38 \text{ J}$$

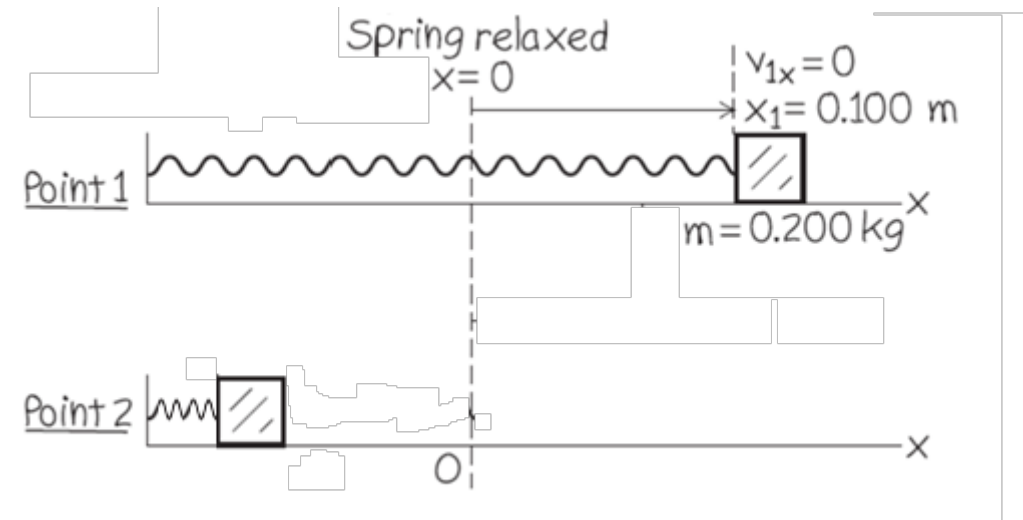
$$v_f = 2.5 \text{ m/s}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.6, p235

Case Problem 4

A glider with mass $m=0.2$ kg sits on a frictionless, horizontal air track, connected to a spring with force constant $k=5.00$ N/m. You pull on the glider, stretching the spring 0.1 m, and release it from rest. The glider moves back toward its equilibrium position ($x=0$). What is its x -velocity when $x= -0.1$ m?



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.7, p239

Case Problem 4: Solution

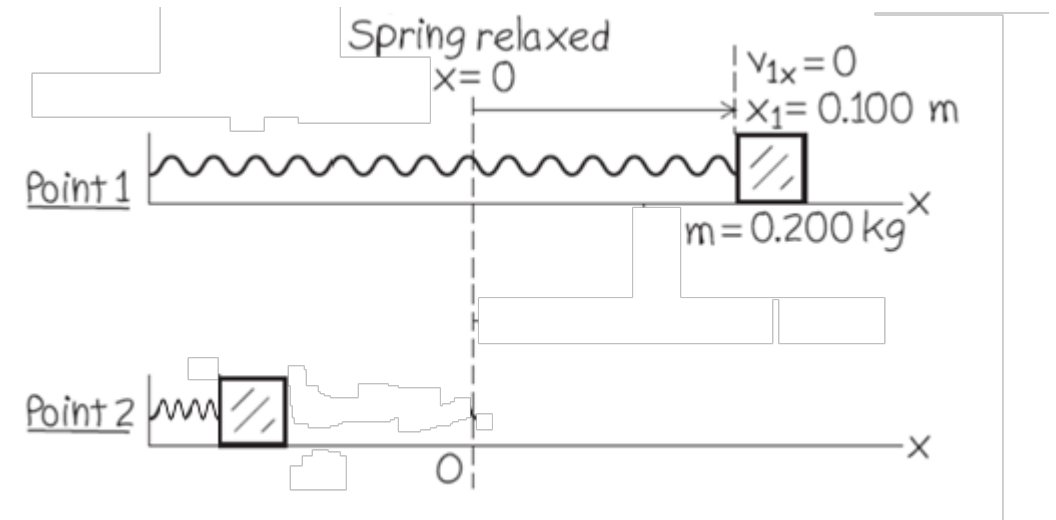
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From conservation of mechanical energy,

$$K_i + U_i = K_f + U_f$$

$$0 + \frac{1}{2} k x_i^2 = \frac{1}{2} m v_f^2 + \frac{1}{2} k x_f^2$$

$$v_f = \sqrt{\frac{k x_i^2 - k x_f^2}{m}} = 0 \text{ m/s}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.7, p239

Case Problem 5

Suppose the glider in Case Problem 4 is initially at rest at $x=0$, with the spring unstretched. You then push on the glider with a constant force with magnitude of 0.61 N in the $+x$ direction. What is the glider's velocity when it has moved to $x=0.1$ m?

Case Problem 5: Solution

Suppose the glider in Case Problem 4 is initially at rest at $x=0$, with the spring unstretched. You then push on the glider with a constant force with magnitude of 0.61 N in the $+x$ direction. What is the glider's velocity when it has moved to $x=0.1$ m?

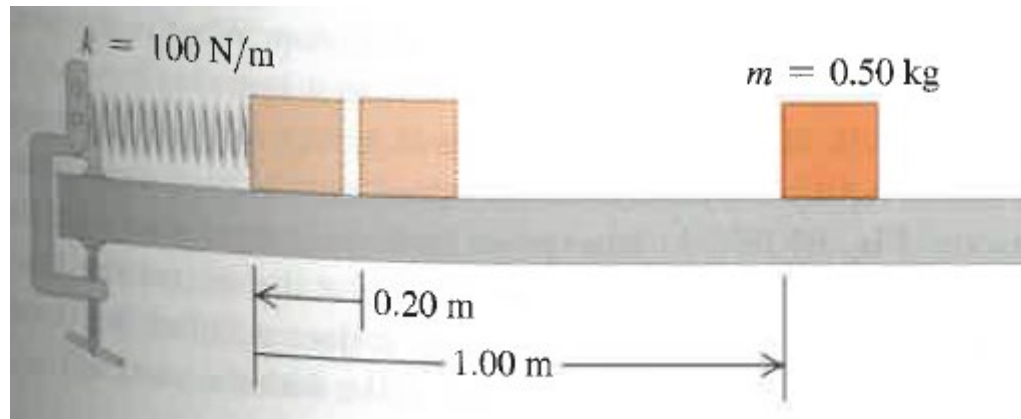
From conservation of energy,

$$\begin{aligned} K_i + U_i + W_{other} &= K_f + U_f \\ 0 + 0 + W_{other} &= \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2 \\ v_f &= \sqrt{\frac{2W_{other} - kx_f^2}{m}} = 0.6 \text{ m/s} \end{aligned}$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Example 7.8, p239

Case Problem 6

A block with mass 0.5 kg is forced against a horizontal spring of negligible mass, compressing the spring a distance of 0.2 m. When released, the block moves on a horizontal tabletop for 1 m before coming to rest. The force constant k is 100 N/m. What is the coefficient of kinetic friction between the block and the tabletop? Take $g = 9.8 \text{ m/s}^2$



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Case Problem 6: Solution

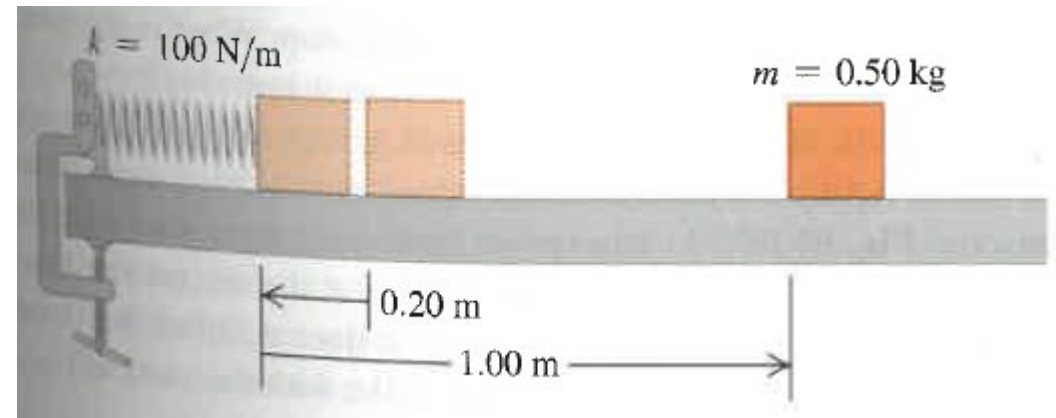
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From conservation of energy,

$$K_i + U_i + W_{other} = K_f + U_f$$

$$0 + \frac{1}{2} k x_i^2 + W_f = 0 + 0$$

$$W_f = -\frac{1}{2} k x_i^2 = -\mu_k m g \Delta x$$



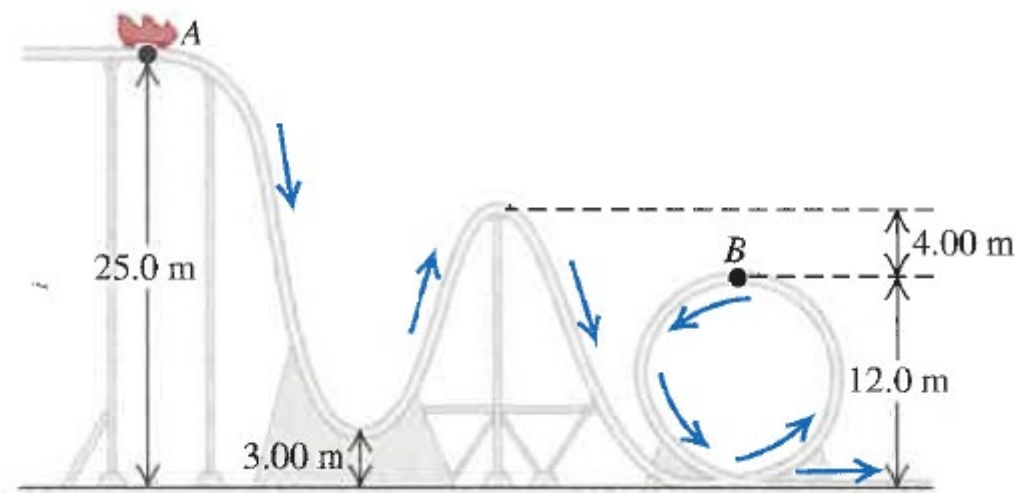
$$\mu_k = \frac{k x_i^2}{2 m g \Delta x} = 0.41$$

H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.39, p255

Off-class practice question(s)

Case Problem 7

A 350-kg roller coaster car starts from rest at point A and slides down a frictionless loop-the-loop. How fast is this roller coaster car moving at point B? Take $g = 9.8 \text{ m/s}^2$



H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.41, p255

Case Problem 7: Solution

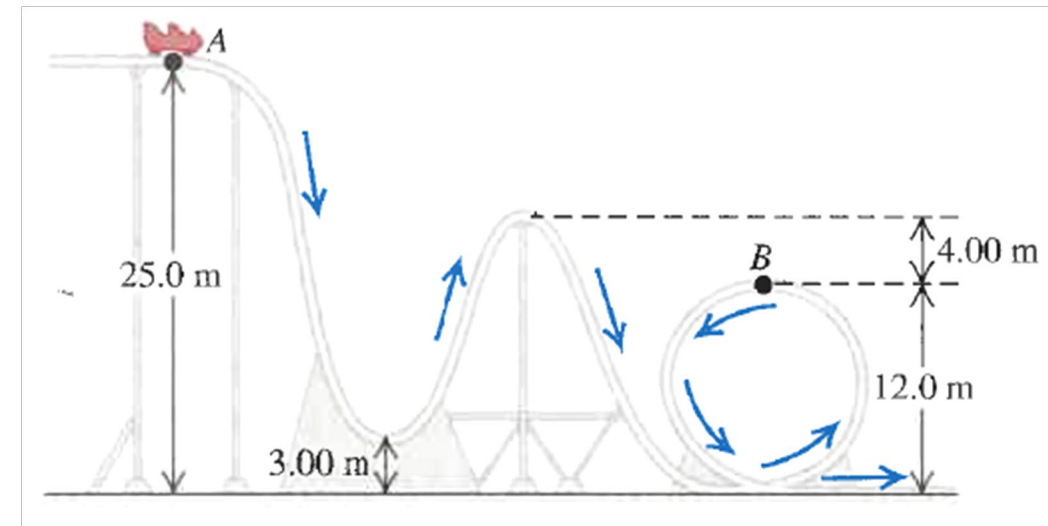
A 350-kg roller coaster car starts from rest at point A and slides down a frictionless loop-the-loop. How fast is this roller coaster car moving at point B?
Take $g = 9.8 \text{ m/s}^2$

From conservation of mechanical energy,

$$K_i + U_i = K_f + U_f$$

$$0 + mgh = \frac{1}{2}mv_f^2 + mg(2R)$$

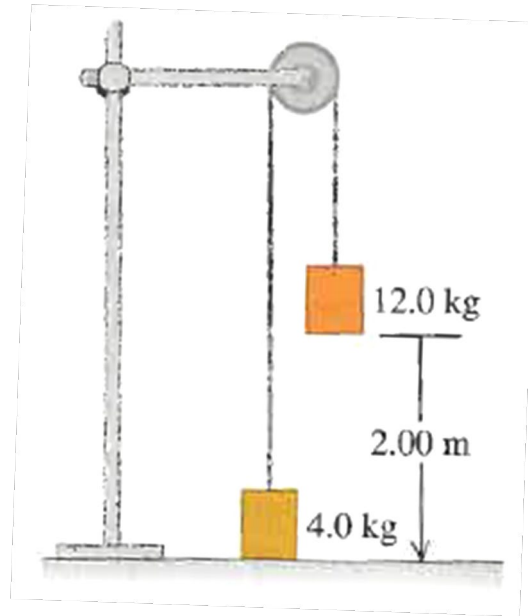
$$v_f = \sqrt{\frac{2mgh - 2mg(2R)}{m}} = 16 \text{ m/s}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.41, p255

Case Problem 8

A system of two paint buckets connected by a light-weight rope is released from rest with the 12-kg bucket 2 m above the floor. Find the speed with which the bucket strikes the floor. Ignore friction and the mass of the pulley. Take $g = 9.8 \text{ m/s}^2$



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Case Problem 8: Solution

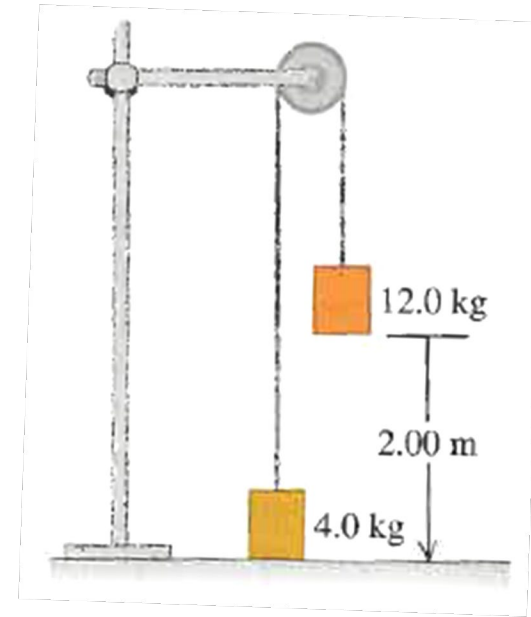
A system of two paint buckets connected by a light-weight rope is released from rest with the 12-kg bucket 2 m above the floor. Find the speed with which the bucket strikes the floor. Ignore friction and the mass of the pulley. Take $g = 9.8 \text{ m/s}^2$

From conservation of mechanical energy,

$$K_i + U_i = K_f + U_f$$

$$0 + m_R gh = \frac{1}{2} (m_L + m_R) v_f^2 + m_L gh$$

$$v_f = \sqrt{\frac{2(m_R gh - m_L gh)}{m_L + m_R}} = 4.4 \text{ m/s}$$



H. D. Young and R. A. Freedman, University Physics (14th edition): Problem 7.51, p256